LESSON 8.2

Write and Solve Equations Using Addition or Subtraction

Set B · Version A · Second form · supported practice

MASTERY CHECK
- ☐ Exceeds
- ☐ Meets target
- ☐ Needs practice

Name _____ **Date** _____ **Period** _____

LEARNING TARGET I can solve one-step addition and subtraction equations using inverse operations.

WHY THIS MATTERS Detective Chen found a coded message: 'The suspect's locker number plus 23 equals 58.' She needs to figure out the locker number to crack the case. The equation is n + 23 = 58. What is n?

REMEMBER

Write and Solve Equations Using Addition or Subtraction

Solving Addition & Subtraction Equations. Formula: x + a = b → x = b − a | x − a = b → x = b + a To solve means to find the value that makes the equation true. I use the inverse operation (the opposite) to get the variable alone, and I do the same thing to both sides.

1. Identify operation on variable.
2. Use inverse operation: subtract to undo addition; add to undo subtraction.
3. Property of Equality: do same operation to both sides.

Words: Equation *(Ecuación)* · Inverse operation *(Operación inversa)* · Isolate *(Despejar)* · Solution *(Solución)* · One-step equation *(Ecuación de un paso)*

1 **Circle the letter of the best answer. Show how you know.**

Which equation represents: 'Three times a number equals 21'?

(A) n + 3 = 21
(B) n − 3 = 21
(C) 3n = 21
(D) n / 3 = 21

SHOW YOUR WORK

2 **Circle the letter of the best answer. Show how you know.**

Which equation represents: 'Seven less than a number is 18'?

(A) 7 − n = 18
(B) n − 7 = 18
(C) n + 7 = 18
(D) n / 7 = 18

SHOW YOUR WORK

3 **Circle the letter of the best answer. Show how you know.**

Which equation represents: 'A number plus 12 equals 30'?

(A) n − 12 = 30
(B) 12n = 30
(C) n + 12 = 30
(D) n / 12 = 30

SHOW YOUR WORK

4 **Circle the letter of the best answer. Show how you know.**

A hiker has walked some distance and still has 4.5 miles left on a 12-mile trail. The equation d + 4.5 = 12 represents her situation, where d is the distance already walked. Solve d + 4.5 = 12.

(A) d = 4.5
(B) d = 7.5
(C) d = 8.5
(D) d = 16.5

SHOW YOUR WORK

5 **Circle the letter of the best answer. Show how you know.**

A hiker has walked some distance and still has 4.5 miles left on a 12-mile trail. The equation d + 4.5 = 12 represents her situation, where d is the distance already walked. Which operation undoes addition?

(A) Multiplication
(B) Division
(C) Subtraction
(D) Squaring

SHOW YOUR WORK

6 **Circle the letter of the best answer. Show how you know.**

A hiker has walked some distance and still has 4.5 miles left on a 12-mile trail. The equation d + 4.5 = 12 represents her situation, where d is the distance already walked. Check: does 7.5 work?

(A) Yes — 7.5 + 4.5 = 12
(B) No — 7.5 + 4.5 = 11
(C) No — 7.5 − 4.5 = 3
(D) Only if the trail is 11 miles

SHOW YOUR WORK

EXPLAIN YOUR THINKING

Why do you have to subtract 19 from BOTH sides, not just one?

SENTENCE STARTER I subtract from both sides because an equation is like a ___ — to keep it balanced, any operation on one side must ___.

3-Point Exit Checklist: ☐ 1. Units & Labels Verified ☐ 2. Mathematical Reasoning Stated ☐ 3. Calculations Double-Checked

How did it go? ☐ 4 · I can teach it ☐ 3 · I've got it ☐ 2 · I need a hint ☐ 1 · I need help

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LEARNING TARGET I can solve one-step addition and subtraction equations using inverse operations.

WHY THIS MATTERS Detective Chen found a coded message: 'The suspect's locker number plus 23 equals 58.' She needs to figure out the locker number to crack the case. The equation is n + 23 = 58. What is n?

PART 1 Understand the idea

models and meaning

1 **Write the letter of the matching item on each line.**

Unlock Card 4: Match each equation to its solution to crack the vault code.

_____ a + 6 = 13	(A) a = 7
_____ b − 5 = 8	(B) b = 13
_____ c + 11 = 20	(C) c = 9
_____ d − 9 = 6	(D) d = 15

PART 2 Apply it

real-world problems

2 **Circle the letter of the best answer. Show how you know.**

Unlock Card 1: Solve x + 7 = 15.

(A) x = 7	(B) x = 8
(C) x = 22	(D) x = 105

SHOW YOUR WORK

3 **Circle the letter of the best answer. Show how you know.**

Unlock Card 2: Solve n − 4 = 9.

(A) n = -5

(B) n = 5

(C) n = 13

(D) n = 36

SHOW YOUR WORK

4 **Circle the letter of the best answer. Show how you know.**

Unlock Card 3: A vault held some clue tokens. After the detective added 16 tokens, there were 50. The equation is t + 16 = 50. How many tokens were there before?

(A) 16

(B) 34

(C) 66

(D) 800

My work — one step per line	What I did to both sides

5 **Circle the letter of the best answer. Show how you know.**

Solve: p + 2.8 = 9.1

(A) p = 3.25

(B) p = 6.3

(C) p = 7.7

(D) p = 11.9

SHOW YOUR WORK

PART 3 Explain and correct

reasoning

6 Find the mistake. Explain it, then show the correct work.

Level 2 Extension — Crack the Detective's Mistake

1. **Equation** x + 8 = 21
2. **Choose inverse** Undo the + 8
3. **Apply to both sides** x + 8 + 8 = 21 + 8
4. **Simplify** x = 29

Which step has the mistake? Explain what went wrong, then show the correct work.

CORRECT WORK

Correct answer: _____

EXPLAIN YOUR THINKING

Why do you have to subtract 19 from BOTH sides, not just one?

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